WashU McKelvey Engineering

Spring 2026 MEMS 3050 Fluid Mechanics and Heat Transfer Laboratory

Lab 5: Heat Exchanger

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Group C (Tuesday 1PM)

I hereby certify that the lab report herein was completed in accordance with the McKelvey School of Engineering and Undergraduate Student academic integrity policies. I attest that artificial intelligence (AI) tools were not used to generate content included in this report.

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ABSTRACT

The objective of the experiment was to evaluate and compare the performances of a shell and tube and a concentric heat exchanger. The apparatus used a hot water supply flowing at 2 L/min and a cold water supply flowing at 4 L/min as the working fluid. As water flowed through the heat exchanger, data was recorded at 20 second intervals for 5 minutes for each device. In the concentric tube heat exchanger, the most heat was transferred in counter flow, with a q_h of 2463.75 W compared to 2351.50 W in parallel flow. In the shell & tube heat exchanger, counter flow was also able to transfer the most heat, with a q_h of 1407.34 W compared to 1281.28 W in parallel flow. It was also found that overall, the concentric tube heat exchanger showcased the highest effectiveness, with an experimental effectiveness of 64.59% for the concentric tube in counter flow and 59.07% in parallel flow. The shell % tube heat exchanger displayed an effectiveness of 37.95% in counter flow and 36.32% in parallel flow. The theoretical effectiveness of the concentric tube heat exchanger was lower than the experimental effectiveness, while the theoretical effectiveness of the shell & tube heat exchanger was higher than the experimental effectiveness.

INTRODUCTION

Heat exchangers are a common device with the goal of changing the temperature of a working fluid. A heat exchanger can either be used to heat a cold fluid to a raised temperature, or to cool and hot fluid to a lower temperature inside of the device. The direction of flow inside the fluid is consequential because it effects the overall effectiveness of the heat exchanger. Two types of flow setups are utilized in this lab, one in which both the hot and cold fluids flow in the same direction, called parallel flow. Initially at the inlet of the heat exchanger in parallel flow there is a large temperature difference between the two fluids, as the hottest and coldest fluids are entering together. Through the heat exchanger the temperatures will move closer together or be the same at the outlet. In the other setup, hot and cold fluids travel in opposite directions, called counterflow. Now, the hot fluid is entering the system with the already warmed cold fluid so the temperature difference is lower than in parallel flow, but this temperature different is maintained across the heat exchanger, so the overall efficiency of the system is higher compared to parallel flow.

In the experiment, two types of heat exchanger were utilized. The first is a concentric tube

heat exchanger, where heat transfer occurs between two tubes, one inside the other, containing fluids of opposite temperature. The other is a shell and tube heat exchanger, where the working fluid runs through many small tubes inside of a large shell containing fluid of the opposite temperature. The shell and tube heat exchanger utilized in this experiment is called a one shell one pass design, where the fluid only has one available path through the device. Optimization of heat exchangers is a common point in many industries requiring changing fluid temperatures, and can involve changes to available heat transfer area through adjustment of pipe diameters, amount of insulation, material properties, flow speeds, and pressures.

In order to perform heat transfer analysis on the system, there are five key assumptions that must be utilized [1]:

(1) The heat exchanger is perfectly insulated, so that $q_c = q_h$.
(2) Axial conduction along the tubes in negligible, because tube walls are assumed to be thin.
(3) Potential and kinetic energy changes are negligible, so there are negligible height or fluid velocity changes throughout the heat exchanger.
(4) Fluid specific heats are constant, so the assumption that $\Delta h = c_p \Delta T$ is valid.
(5) The overall heat transfer coefficient U is constant.

The shell and tube heat exchanger is simplified for the purposes of calculation as two rectangles for each fluid that experience heat transfer through a heat transfer surface area, A_s. The total heat transfer in both heat exchangers can then be represented as Eq. 1,

$$q = UA(T_H - T_C) \tag{1}$$

Where q is the total heat transfer in the heat exchanger in W, U is the overall heat transfer coefficient of the heat exchanger in $\frac{\mathrm{W}}{\mathrm{m}^2 \cdot \mathrm{K}}$, A is the total heat transfer surface area in m^2. For a concentric tube heat exchanger, $A = \pi d L$, where d is the outer diameter of the inner tube and L is the length, both in m. For a shell and tube heat exchanger, $A = N\pi d L$, where N is the number of tubes. T_H and T_C are the temperature of the hot fluid and cold fluid respectively in °C. Because these temperatures will vary greatly along the length of the heat exchanger as energy is transferred, the ΔT to represent the

total heat transfer in the heat exchanger is the log-mean temperature difference, shown in Eq. 2,

$$\Delta T_{lm} = \frac{\Delta T_2 - \Delta T_1}{\ln(\frac{\Delta T_2}{\Delta T_1})} \tag{2}$$

In parallel flow, $\Delta T_1 \equiv T_{h,i} - T_{c,i}$ and $\Delta T_2 \equiv T_{h,o} - T_{c,o}$. In counter flow, one of the fluids moves in the opposite direction, so $\Delta T_1 \equiv T_{h,i} - T_{c,o}$ and $\Delta T_2 \equiv T_{h,o} - T_{c,i}$, all in K. Eq. 1 can then be rewritten as

$$q = UA\Delta T_{lm} \tag{3}$$

Which can be rearranged to find the overall heat transfer coefficient. The heat gained in W by the cold fluid in the heat exchanger, q_c, can be calculated using Eq. 4,

$$q_c = (\dot{m}c_p)_c(T_{c,o} - T_{c,i}) \tag{4}$$

Where $\dot{m}$ is the mass flow rate of the cold fluid in $\frac{\text{kg}}{\text{s}}$, c_p is the specific heat capacity of the cold fluid in $\frac{\text{kJ}}{\text{kg·K}}$, $T_{c,o}$ is the temperature in K of the cold fluid outlet, and $T_{c,i}$ is the temperature in K of the cold fluid inlet. Similarly, the heat lost in W by the hot fluid in the heat exchanger, q_h, can be calculated using Eq. 5,

$$q_h = (\dot{m}c_p)_h(T_{h,i} - T_{h,o}) \tag{5}$$

Where now $\dot{m}$, c_p, $T_{h,i}$, and $T_{h,o}$ are parameters of the hot fluid. From the first assumption made to perform heat transfer analysis, in a perfectly insulated heat exchanger the difference between q_c and q_h will be 0. However, in a real heat exchanger there are losses to the environment from imperfect insulation. Therefore, the heat loss to the environment in W, q_L, can be calculated using Eq. 6,

$$q_L = q_h - q_c \tag{6}$$

The effectiveness of a heat exchanger can be found by taking the ratio between the total heat transfer q and the maximum possible heat transfer, $q_{\max}$. The maximum possible heat transfer can be found using Eq. 7,

$$q_{\max} = C_{\min}(T_{h,i} - T_{c,i}) \tag{7}$$

$C_{\min}$ represents the minimum heat capacity rate in $\frac{\text{W}}{\text{K}}$, and is equivalent to the lowest $\dot{m}c_p$ of the two fluids. The minimum heat capacity rate limits how much energy can be exchanged. Similarly, $T_{h,i} - T_{c,i}$

is the maximum possible temperature difference in the heat exchanger. The effectiveness is then equal to Eq. 8,

$$\varepsilon = \frac{q}{q_{\max}} \tag{8}$$

This effectiveness is to be compared to the theoretical effectiveness, which in parallel flow is calculated by Eq. 9:

$$\varepsilon_{\text{theoretical,parallel}} = \frac{1 - \exp[-\text{NTU}(1 + C_r)]}{1 + C_r} \tag{9}$$

Where the number of heat transfer units (NTU) can be found using Eq. 10,

$$\text{NTU} = \frac{UA}{C_{\min}} \tag{10}$$

Where A is the outer diameter of the inner tubes in m^2. For the case of counter flow, the theoretical effectiveness is found using Eq. 11:

$$\varepsilon_{\text{theoretical,counter}} = \frac{1 - \exp[-\text{NTU}(1 - \text{C}_\text{r})]}{1 - C_r[-\text{NTU}(1 - C_r)]} \tag{11}$$

METHODS

Apparatus. Table 1 below lists the equipment used in the lab.

Table 1

Equipment	Quantity	Description
Concentric Tube Heat Exchanger	1	EDIBON Extended Concentric Tube Heat Exchanger, "TITCA"
Shell & Tube Heat Exchanger	1	EDIBON Shell & Tube Heat Exchanger, "TICT"
Control Interface Box	1	EDIBON Control Interface Box, with SCADA and PID Control
Base Service Unit	1	EDIBON Base Service Unit
Computer	1	Used with EDIBON control software

The concentric and shell & tube heat exchangers were used as the systems being studied in the experiment. The control interface box was the connection point of all of the built in sensors and the

computer software, as well as the main source of power for the system. The base service unit supplied the hot water from a built in reservoir and the cold water through a connection point to the wall supply.

Figure 1 shows a schematic of the shell % tube heat exchanger utilized in the lab.

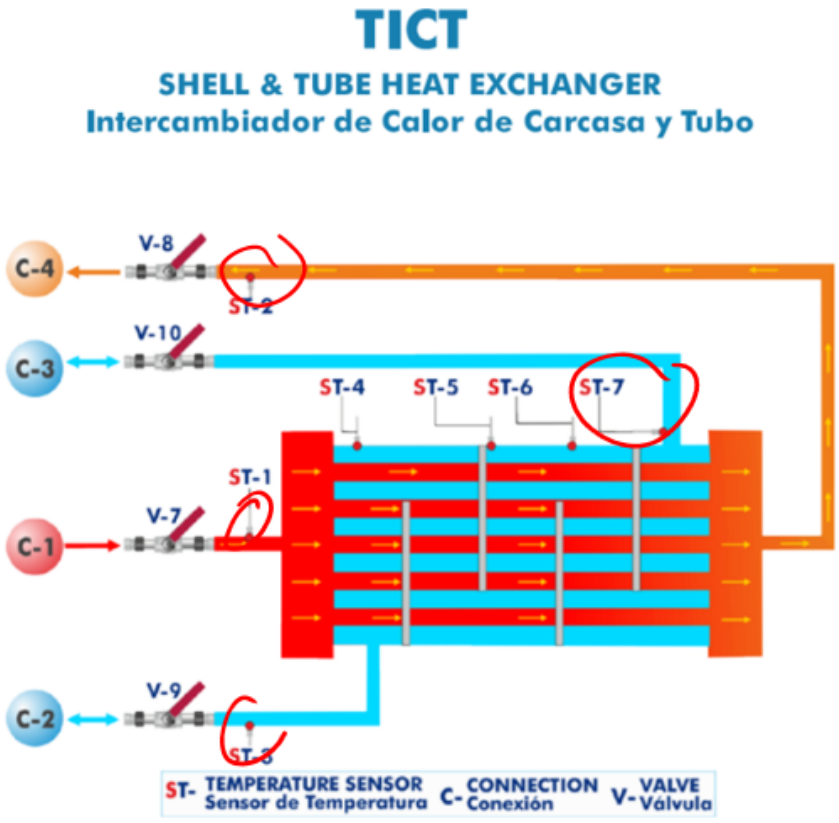


Figure 1 Schematic of the Shell & Tube Heat Exchanger

In the experiment, the inlet and outlet thermocouples are marked with the red circles, with the hot fluid being recorded by ST-1 and ST-2, while the cold fluid was recorded by ST-3 and ST-7. A system of 21 tubes were included inside of the shell.

Figure 2 shows a schematic of the concentric tube heat exchanger utilized in the lab.

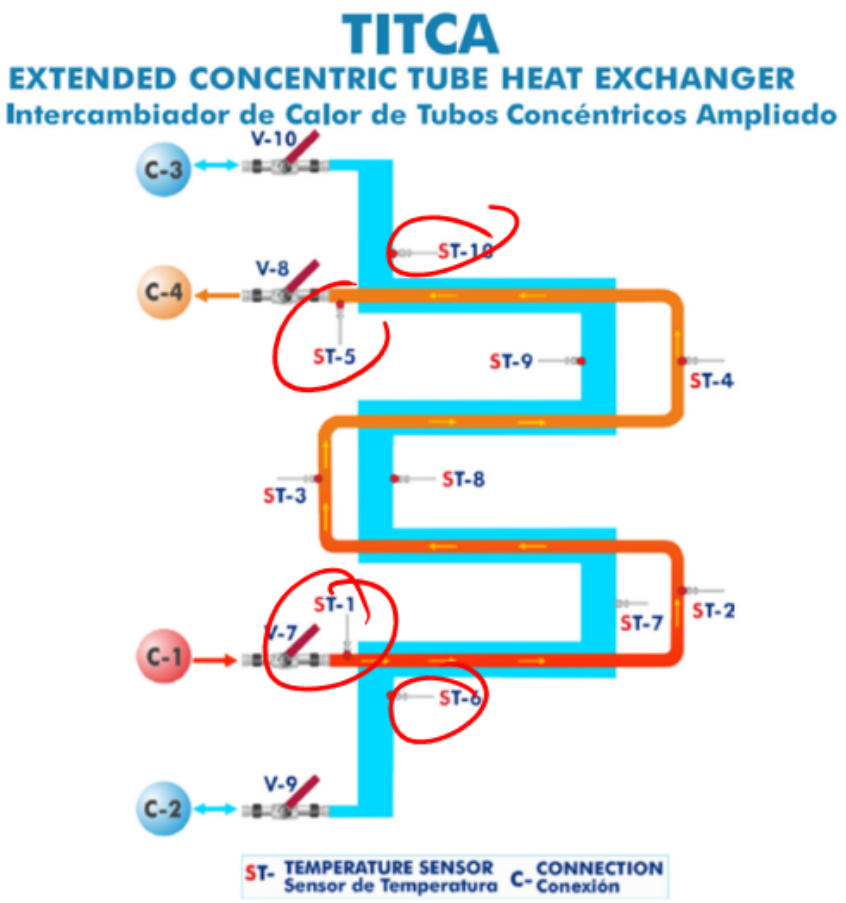


Figure 2 Schematic of the Concentric Tube Heat Exchanger

In the experiment, the inlet and outlet thermocouples are once again marked with red circle. The hot fluid was recorded by ST-1 and ST-5, while the cold fluid was recorded by ST-6 and ST-10.

Procedure. Before experiments could begin, the hot water supply was heated from a starting environmental temperature to 65°C. Once this temperature was reached, it was then necessary to purge the air in the heat exchanger using the cold water supply and opening the valves located on lines feeding out of the heat exchanger. It was observed that the lines pushed air out of the heat exchanger, indicating that the cold water was at a high enough flow rate. Using the computer software, the flow rate of the hot water was set to $2\frac{\text{L}}{\text{min}}$. Using the valve attached to the cold water supply, the flow rate of the cold water was set to $4\frac{\text{L}}{\text{min}}$.

Thermocouples along the length of the heat exchanger was used to record temperatures at each location using the computer software. Once the hot and cold water loops were turned on, system was left running until thermocouple temperatures reached steady state, observed on the software using the

built in temperature graphs. Data was then recorded at steady state every 20 seconds for both parallel and counter flow for both the concentric and shell and tube heat exchangers.

Analysis. Experimental data was collected and analyzed using Excel and MATLAB. Excel was used for the calculation of the values in Table 5 using the equations written in the Introduction section. MATLAB was used to plot the temperature versus position for the concentric tube heat exchanger using data imported from the computer software. A simple error analysis was performed to compare the theoretical and experimental effectiveness values in each heat exchanger case.

Possible sources of error in the experiment come from the five key assumptions made in order to calculate heat transfer parameters and the overall procedure of the experiment. It was assumed that axial conduction along the tubes was negligible because tube walls were assumed to be thin, however in the real system there are losses due to axial conduction not reflected in the calculations. Throughout the heat exchanger, especially for the shell and tube case, it was assumed that there was negligible height or fluid velocity changes throughout the heat exchanger, but a variety of factors including flow speed, direction, and temperature would cause variation in the potential and kinetic energy of the fluid. It was also assumed that the fluid specific heats were constant, when in reality this and other fluid properties vary with temperature. Finally, it was also assumed that the overall heat transfer coefficient was constant, but this also varies with temperature of the fluid.

Sources of error in the overall procedure come from not waiting until the fluid had reached complete steady state before data collection, not ensuring that valves were completely open or closed during experimental operation, needing to add more water to the hot water supply mid-experiment, errors in the thermocouple temperature recording, and rounding during heat transfer calculations that may have propogated error throughout the analytical results.

RESULTS & DISCUSSION

Figure 3 shows thermocouple temperature versus position for the concentric tube heat exchanger running in parallel flow.

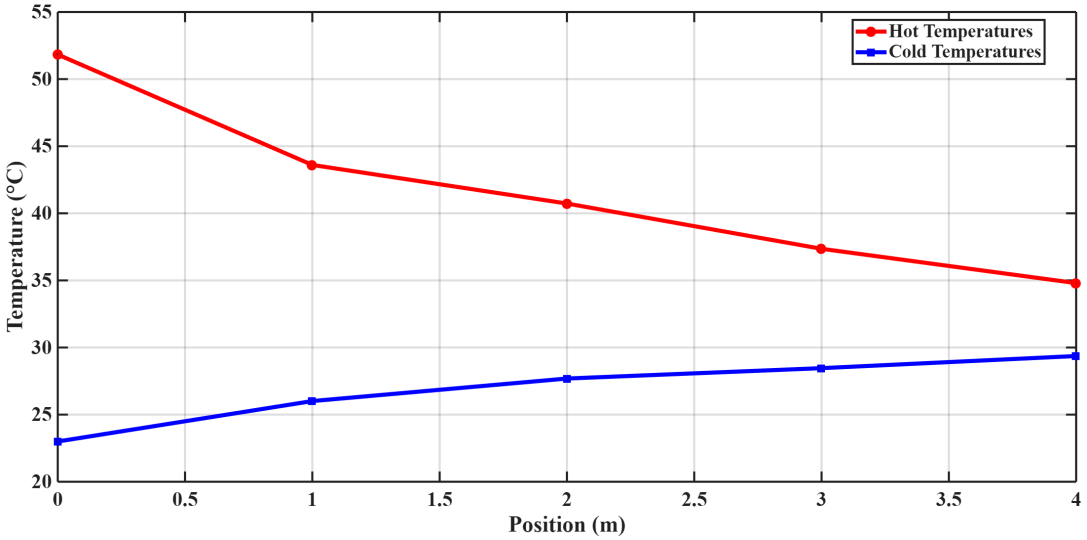


Figure 3 Concentric Heat Exchanger Temperature versus Position in Parallel Flow

At x = 0, the hot and cold water is at the inlet, which is the location of the greatest difference in temperature, ΔT = 51.8325 – 22.9973 = 28.8352 °C. This is therefore also where the greatest amount of heat transfer occurs, which can be seen from the large drop in the temperature of the hot fluid as it moves through the heat exchanger. However, consistent with theory behind parallel flow, the temperatures of the two fluids begin the move closer which causes the heat transfer rate to decrease. The hot and cold fluid temperatures begin to level off, until they reach the outlet, which is the location of the lowest difference in temperature, ΔT = 34.7997 – 29.3663 = 5.4334 °C. In parallel flow, outlet temperatures will approach one another and eventually will reach the same temperature, but cannot overlap. The results of the experiment in Fig. 3 are consistent with that theory.

Figure 4 shows thermocouple temperature versus position for the concentric tube heat exchanger running in counter flow.

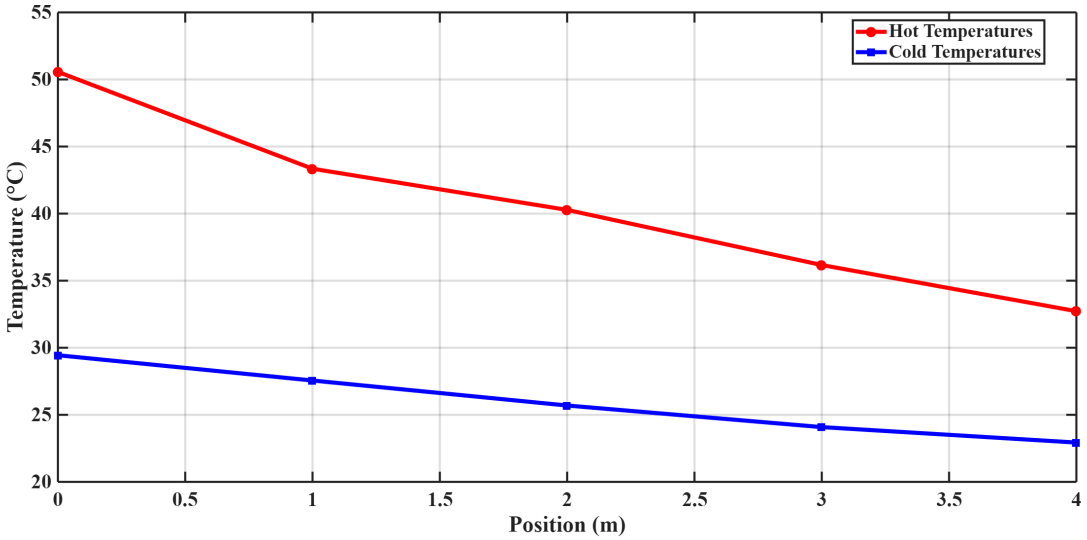


Figure 4 Concentric Heat Exchanger Temperature versus Position in Counter Flow

In the counter flow arrangement, the hot and cold water are now moving in opposite directions, with the cold water entering at x = 4m with the exiting hot water. The highest temperature difference is at x = 0, with a value of ΔT = 50.6544 – 29.4419 = 21.1226 °C. This is lower than the highest temperature difference in the parallel flow case, consistent with the theory that parallel flow maximizes this temperature difference at the inlets. However, it can be seen in Fig. 4 that the overall temperature difference remains much more consistent than in the parallel case, indicating that there is consistently a higher amount of heat transfer occurring between the two fluids. This increased amount of heat transfer across the system demonstrates why running fluids in counter flow for a concentric tube heat exchanger can increase the overall effectiveness of the system.

Table 5 summarizes the heat transfer calculations made for each heat exchanger type in each flow configuration, using the equations laid out in the Introduction section.

Figure 5 Summary of Heat Transfer Calculations

Heat Exchanger	Flow Configuration	q_h [W]	q_c [W]	q_L [W]	Effectiveness	dT_lm [K]	U [W/m^2*K]	NTU	Theoretical Effectiveness
Concentric Tube	Parallel	2351.5036	1769.0994	582.4041	0.5907	14.0212	741.4427	1.2148	0.5596
	Counter	2463.7479	1806.8285	656.9194	0.6459	14.7316	739.3739	1.2114	0.6252
Shell and Tube	Parallel	1281.2752	1428.0367	-146.7615	0.3632	17.3531	223.8346	0.5348	0.3680
	Counter	1407.4636	2483.3432	-1075.8796	0.3795	17.2852	246.8448	0.5898	0.4071

For both heat exchangers, it can be seen by looking at both q_h and q_c that there is more heat transfer occurring in the counter flow configuration compared to the parallel flow configuration. The lost by the counter flow in the concentric case was 112.2443 W higher than that of the parallel flow, and in the shell and tube case the heat lost in counter flow was 126.1884 W higher than that of the parallel flow. The same is true for the effectiveness, which is also higher for both heat exchangers in counter flow. This aligns with the hypothesis that the consistently large ΔT across the heat exchanger in counter flow outweighs the initially high ΔT in parallel flow that causes heat transfer to occur rapidly at the inlet but quickly dampen by the outlet. In counter flow, the heat transfer is able to remain high throughout the process because of the small difference in ΔT across the system.

The most effective heat exchanger was the concentric tube, shown by its maximum effectiveness of 0.6459 when running in counter flow. The least effective was the shell and tube heat exchanger running in parallel flow, which only had an effectiveness of 0.3632. While the concentric tube heat exchanger was more effective, the shell and tube heat exchanger had a greater average temperature, shown by the log-mean temperature in Table 5. The reasoning for the lower effectiveness could be that the conditions of the experiment were not suited for a shell and tube design, as their increased heat transfer surface area from the number of tubes should increase its overall effectiveness. Shell and tube designs are also made to be used in cases of very high temperatures and pressures, so it is possible that the experiment did not reach those high levels where it may perform better. Another interesting observation from the experiment is that in the concentric tube heat exchanger, switching from parallel to counter flow caused an increase in effectiveness of 5.52%, while in the shell and tube heat exchanger the effectiveness only rose by 1.63%. Counter flow can then be attributed to be more crucial for effectiveness in the case of a concentric tube heat exchanger, while in a shell and tube design effectiveness is not as dependent on the direction of flow.

The percent error was calculated between the experimental and theoretical effectiveness in each case. For the concentric tube heat exchanger, the percent error in parallel flow was 5.56%, and the percent error in counter flow was 3.31%. For the shell & tube heat exchanger, the percent error in parallel flow was 0.48%, and the percent error in counter flow was 2.76%. It can be observed from Table 5 that the experimental effectiveness in the concentric tube heat exchanger was higher than the theoretical effectiveness for both flow configurations.

Another interesting observation is that in the shell & tube heat exchanger the heat gained by the cold fluid is consistently higher than the heat lost by the hot fluid; in counter flow the cold fluid gained over 1000 W more heat than the hot fluid lost. This indicates that the cold fluid was able to receive much more heat than the hot fluid lost. Possible explanations for this could be that the method for heat transfer calculations in the shell & tube heat exchanger did not accurately reflect the experimental model, or that there was a mistake made during the experiment that led to this error.

The reasoning for unexpected final results can be attributed to many possible sources of uncertainty in our experiment. For one, the thermocouples used in the heat exchangers only measure data at a singular location, but their data is used to make a generalization for the temperature across a longer span. This can make the temperature values differ from what may have really occurred inside the system. There is also error propagated through the results based on the ability of the sensors to make fine tuned measurements, and some error should be attributed to each data point recorded by the sensors in the heat exchangers. There is also uncertainty in the consistency of the flow rate measurement, as in the calculations it was taken to be a constant 2 and 4 L/min, but in reality the 4 L/min was not constant and was different in each case, as the physical valve had to be opened by hand until it was approximately the desired value. Another source of uncertainty was that the evaluation of steady-state was done by the human eye, which is likely not exactly where the fluid reached steady-state. Results were taken for 500 seconds and averages were taken, but it is likely there is some error from the fact that the fluid was still changing temperature during the recording of data. Finally, the quality of the thermocouples and their ability to register the correct temperatures and report it back to the software correctly also carries some uncertainty, especially between the two different heat exchangers.

Finally, there is uncertainty in the comparison between the real world and the assumptions required in the Introduction section in order to calculate the heat transfer properties in Table 5. For one, it was assumed that the heat exchanger was perfectly insulated, but it can be seen that there are actually significant losses that occurred in each configuration. Second, we assumed that there was negligible axial conduction along the tubes, but in reality there will be additional losses due to heat being transferred along the length of the tubes. Third, we assumed that there was negligible potential and kinetic energy changes, but the geometry of the heat exchangers plays a significant role in creating

potential and kinetic energy changes. In the concentric heat exchanger, the bends in the tubes would make non-negligible changes to recorded values, and in the shell & tube the tubes cause the flow path of the water to change. We also assumed that the properties of the fluid such as C_p, ρ, and U were constant, but these values vary with temperature.

CONCLUSION

The lab was successfully able to compare the performance of a concentric tube and a shell & tube heat exchanger through the process of parallel and counter flow studies. Through the experiment, it was found that under the prescribed conditions the concentric tube heat exchanger in counter flow was able to transfer the most heat of 2463.75 W and had the highest experimental effectiveness of 64.59%. The lowest heat transfer occurred in the case of the shell & tube heat exchanger running in parallel flow, which was 1281.28 W, and also had the lowest experimental effectiveness of 36.32%. The experiment supported heat exchanger theory by showcasing the increased ability of counter flow heat exchangers to transfer heat through higher heat transfer and effectiveness. An error analysis was performed between the experimental and theoretical effectiveness, and it was found that there was below a 6% error between the two methods of determination. This error can be attributed to the assumptions required to utilize the equations listed in the Introduction section, and heat losses to the environment.

A shortcoming of the experiment might be that losses to the environment were calculated, but were not later utilized to effect the conclusions of the heat transfer or the effectiveness of each setup. Due to the variance in the amount of losses in each setup, further analysis could be done to determine the causation of increased losses between concentric and the shell & tube exchanger.

References

[1] Whalen, E., 2026, "LAB 5: HEAT EXCHANGERS," Washington University in St. Louis, accessed 3/24/2026, https://wustl.instructure.com/courses/173071/files?preview=10953322